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Small volume processing systems and methods with capacitive sensing

Abstract

A fluid processing system may include a flow control cassette comprising at least one interface sensor chamber in fluid communication with at least one of a plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber. The fluid processing system may include, in the alternative or in addition, at least one syringe comprising a wall defining a barrel having a first end and a second end, the barrel having a bore with or without a piston or plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent App. No. 62/925,303, filed Oct. 24, 2019, which is expressly incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure is generally directed to systems and methods for processing (e.g., washing) suspensions of cells. More particularly, the present disclosure is directed to systems and methods for processing small volumes of cells using a disposable fluid circuit and a reusable processing machine or hardware to generate a cell product.

BACKGROUND

(2) A number of well-known therapies are currently practiced in which a targeted cellular blood component (e.g., red blood cells, white blood cells, and platelets) is separated from whole blood and stored for later infusion to a patient. The targeted cell product (e.g., red blood cells, white blood cells, or platelets) may be in a suspension that includes plasma and/or some other supernatant. As such, it is sometimes desirable to “wash” the cellular suspension (typically with a physiologic buffer) to remove the plasma/supernatant, as well as any non-target cellular material, prior to reinfusion.

(3) Systems and methods for cell washing are exemplified by US Pub. Nos. 2013/0341291, 2013/0092630, and 2014/0199680, each of which is incorporated herein by reference. Each of these published applications discloses cell washing methods utilizing disposable fluid circuits including a spinning membrane separator and a reusable processing machine. Such machines include peristaltic pumps and pinch valves that act on the tubing of the fluid circuit to direct flow within the fluid circuit.

(4) The fluid circuits in the published applications listed above have a relatively large internal volume, and thus require relatively large volumes of wash or flush media to clear processed fluid through the fluid circuit. While such systems and fluid circuits are capable of washing and reducing the volume of the targeted cell component into final volumes of ranging from approximately 50 mL to 5,000 mL, there are instances in which smaller final volumes (e.g., 10 mL) are desired, such as when processing single-dose quantities of mononuclear cell products. Thus, it would be desirable to provide systems and methods for processing (e.g., concentrating or washing) small volumes of cellular suspensions.

SUMMARY

(5) In an aspect, a fluid processing system includes a disposable fluid circuit and reusable hardware configured to accept the disposable fluid circuit. The disposable fluid circuit includes a separation chamber and a flow control cassette. The flow control cassette includes a housing containing a

plurality of separate channels connected to a plurality of selectable junctions, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber. The reusable hardware includes a separator connected to the separation chamber, a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor, and at least one controller coupled to the separator, the at least one actuator and the capacitive sensor via the coupling, the controller configured to selectively operate the separator and the at least one actuator to provide a procedure according to a protocol.

(6) In another aspect, a fluid processing system includes a disposable fluid circuit and reusable hardware configured to accept the disposable fluid circuit. The disposable fluid circuit includes a separation chamber, at least one plungerless syringe, the syringe including a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe, and a flow control cassette including a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe. The reusable hardware includes a separator connected to the separation chamber, at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe, a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor, and at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the controller configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

(7) In a further aspect, a fluid processing system includes a disposable fluid circuit and reusable hardware configured to accept the disposable fluid circuit. The disposable fluid circuit includes a separation chamber; at least one plungerless syringe, the syringe including a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe, and a flow control cassette. The flow control cassette includes a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber. The reusable hardware includes a separator connected to the separation chamber, at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe, a coupling connected to the at least one capacitive sensor disposed on an outer surface of the syringe, a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber, and at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the controller configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a system for processing (e.g., concentrating or washing) small volumes of cellular suspensions including a disposable fluid circuit and a reusable processing machine or hardware.
- (2) FIG. 2 is a plan view of an embodiment of a disposable fluid circuit for use in the system of FIG. 1.
- (3) FIG. 3 is a block diagram of an embodiment of the reusable processing machine or hardware of FIG. 1.
- (4) FIG. 4 is a side view of an embodiment of a flow control cassette for use in the disposable fluid circuit of FIG. 2, with one of the sidewalls removed for ease of illustration.
- (5) FIG. 5 is a side view of the embodiment of the flow control cassette of FIG. 4, illustrating the opposite side of the flow control cassette.
- (6) FIG. 6 is a side view of the embodiment of the flow control cassette of FIG. 4, with the location of the selectable junctions, or clamps, from the opposite side of the cassette illustrated relative to the channels and apertures illustrated in FIG. 4.
- (7) FIG. 7 is a cross-sectional view of one of the clamps of the flow control cassette being manipulated by an actuator.
- (8) FIG. 8 is a plan view of an embodiment of a flow control interface that is associated with the flow control cassette, and which is configured to operate the selectable junctions, or clamps, of the flow control cassette.
- (9) FIG. 9 is a side view of another embodiment of a flow control cassette for use in the disposable fluid circuit of FIG. 2, the cassette incorporating capacitive sensors for use with air sensor chambers;
- (10) FIG. 10 is a perspective view of a separation/washing device using a spinning membrane.
- (11) FIG. 11 is a cross-sectional view of an embodiment of a syringe/syringe pump as may be used as part of the system of FIG. 1, with a piston head assembly in a first position.
- (12) FIG. 12 is a cross-sectional view of the syringe/syringe pump of FIG. 11, with a piston head assembly in a second position.
- (13) FIG. 13 is a chart of the signal response of one of the infrared detectors or sensors of the syringe/syringe pump of FIG. 11.
- (14) FIG. 14 is a side view of another embodiment of a syringe/syringe pump as may be used as part of the system of FIG. 1, with a capacitive sensor applied to an outer surface of the syringe.
- (15) FIG. 15 is a cross-sectional view of the syringe/syringe pump of FIG. 14, with an air/fluid interface in a first position.
- (16) FIG. 16 is a cross-sectional view of the syringe/syringe pump of FIG. 14, with an air/fluid interface in a second position.
- (17) FIG. 17 is a flowchart illustrating an embodiment of a method of operating the system of FIG. 1.
- (18) FIG. 18 is a flowchart illustrating an embodiment of a procedure that may be performed according to a protocol using the system 100.

DETAILED DESCRIPTION

- (19) A more detailed description of the systems and methods in accordance with the present disclosure is set forth below. It should be understood that the description below of specific devices and methods is intended to be exemplary, and not exhaustive of all possible variations or applications. Thus, the scope of the disclosure is not intended to be limiting, and should be understood to encompass variations or embodiments that would occur to persons of ordinary skill.
- (20) Turning first to FIG. 1, an embodiment of a system 100 for processing fluids, such as cell suspensions (e.g., cell washing), is illustrated, the system 100 including a disposable fluid circuit (also referred to as a set or kit) 102 and a reusable processing machine, or hardware, 104.

(21) As seen in FIGS. 1 and 2, the disposable fluid circuit **102** is connectable to a source container **110** of fluid, in particular biological fluid. The disposable fluid circuit **102** includes a separator chamber, in this case defined by a spinning membrane separator **112** that is used to process the fluid received from the source container **110**, and to direct a portion of that fluid into one of more product containers **114**. These containers may be in the form of flexible bags according to the illustrated embodiment. The flow of fluid from the source container **110**, through the spinning membrane separator **112**, and to the one or more product containers **114** is achieved through the use of first and second syringes **116**, **118**, which are in fluid communication with the source container **110**, the spinning membrane separator (or spinning membrane for short) **112**, and the one or more product containers **114**. The syringes **116**, **118** also may be in fluid communication with a number of other containers **120**, **122**, **124**, **126**, **128** (container **124** being schematically represented in FIG. 2 only).

(22) The flow of the fluid between the containers **110**, **114**, **120**, **122**, **124**, **126**, **128**, the spinning membrane **112**, and the syringes **116**, **118** is controlled using a flow control cassette **130**, which cassette **130** may be connected to each of the foregoing by tubing, or lines. In addition, the cassette **130** may include internal flow paths that are defined in part by a plurality of separate channels or passages, which in turn may be contained within the structure (e.g., housing) of the cassette **130**. The channels may be defined by the structure (e.g., housing) of the cassette **130**, as illustrated, or may be defined by tubing or lines disposed within the cassette, see U.S. Pub. No. 2017/0204371, which is incorporated herein by reference. The channels may be connected at a plurality of selectable junctions, which may control the flow of fluid from one channel to another. These selectable junctions may also be referred to as valves, valve stations, or clamps, because, as illustrated, the selectable junctions provide controlled access between the channels. The cassette **130** may also include sensor stations, by which sensors may be associated with the flow paths within the cassette **130** to determine characteristics of the flow therein, such as pressure, presence of air and/or fluid, or optical properties. Preferably, the length of each of the lines and channels is kept as short as possible to further minimize the internal volume of the fluid circuit **102**.

(23) As illustrated in FIG. 2, the spinning membrane **112** and the syringes **116**, **118** may be integrally formed as part of (i.e., as one piece with) the cassette **130**, so as to further reduce the tubing volume associated with the kit **14**. According to other embodiments, the spinning membrane **112** and/or the syringes **116**, **118** may be attached to the remainder of the fluid circuit **102** at the time of use, as may be the case with one or more of the containers **110**, **114**, **120**, **122**, **124**, **126**, **128**. Again, as illustrated in FIG. 2, the container **120** and container **128** are integrally formed with the cassette **130**.

(24) As seen in FIGS. 1 and 3, the reusable hardware component (or reusable hardware for short) **104** includes a drive **140** for the spinning membrane separator **112**, a syringe pump **142**, **144** for each syringe **116**, **118**, and a control cassette interface **146** that is associated with the flow control cassette **130** when the fluid circuit **102** is disposed on the hardware **104** (e.g., is mounted on the hardware **104**). As will be explained in detail below, the cassette interface **146** includes actuators and sensors that are associated with the clamps and sensor stations of the flow control cassette **130**, and are configured to operate the clamps or sense characteristics of the fluid, respectively.

(25) The reusable hardware **104** also includes a controller **150** that is configured to control operation of the system **100**, for example using a method of operation as is explained below relative to FIGS. 17 and 18. The controller **150** may include a microprocessor **152** (which, in fact may include multiple physical and/or virtual processors). According to other embodiments, the controller **150** may include one or more electrical circuits designed to carry out the actions described herein. In fact, the controller **150** may include a microprocessor **152** and other circuits or circuitry. In addition, the controller **150** may include one or more memories **154**. The instructions by which the microprocessor **150** is programmed may be stored on the one or more memories **154** associated with the microprocessor **150**, which memory/memories **154** may include one or more

tangible non-transitory computer readable memories, having computer executable instructions stored thereon, which when executed by the microprocessor **152**, may cause the microprocessor **152** to carry out one or more actions as described below.

(26) The controller **150** may be coupled (i.e., directly or indirectly connected) to the equipment of the reusable hardware **104**, such as the spinning membrane drive **140**, the first syringe pump **142**, the second syringe pump **144**, and the cassette interface **146**. The controller **150** may operate each of these devices, each of which may be an assembly of other devices or equipment, to cause the fluid to flow through the fluid circuit **102** associated with the hardware **104**, for example to cause fluid to flow from the source container **110**, through the spinning membrane **112**, and eventually into the product container(s) **114**. For example, the controller **150** may be programmed to perform a process or procedure according to a protocol, such as to wash particular cells contained in the fluid within the source container **110**, before they are distributed into one or more of the product containers **114**. The controller **150** may be programmed to perform other actions as well, such as to test the fluid circuit **102**, to prime the fluid circuit **102**, to rinse parts of the circuit **102** after the wash has been performed, to add other components to the cell-containing fluid before that fluid is distributed to the product containers **114**, and to distribute the cell-containing fluid into the product containers **114**.

(27) Having thus described the structure and operation of the system **100**, including the fluid circuit **102** and reusable hardware **104**, in general terms, the details of each of the systems is now discussed, starting with the fluid circuit **102**.

(28) As mentioned above, the flow of fluids through the fluid circuit **102** is controlled through the flow control cassette **130**. While other embodiments may involve fluid circuits **102** where some of the fluid does not pass through the cassette **130**, according to the illustrated embodiment, the fluid flows between the containers **110**, **114**, **120**, **122**, **124**, **126**, **128**, the spinning membrane **112**, and the syringes **116**, **118** via the cassette **130**. As mentioned above, each of the containers **110**, **114**, **120**, **122**, **124**, **126**, **128**, the spinning membrane **112**, and the syringes **116**, **118** is connected to the cassette through the use of medical grade tubing, or lines.

(29) With reference to FIGS. **1** and **2**, the container **120** used to receive the filtrate of the spinning membrane **112** and other fluids is connected via a line **162** to a filtrate container port **164** formed on the cassette **130**. The first and second containers **122**, **124**, used to contain wash media as may be used during the method of operation of the system **100**, each may be connected to a line **166**, **168** that are connected at a first end to the containers **122**, **124**, and at a second end to a Y-junction **170**. The Y-junction **170** is, in turn, connected via a line **172** to a wash container port **174**. The container **126**, which may contain a cryopreservation agent (CPA) according to one embodiment, is connected via a line **176** to a port **178**. The source container **110** may be connected via a line **180** to a source container port **182**. Further, a secondary container **128** is connected via a line **184** to a port **186**, and the product container(s) **114** is/are connected via a line **188** to a product container port **190**.

(30) As is reflected in the illustrated embodiment, certain of the containers may be formed integrally with the fluid circuit **102**, while other containers may be attached at the time of operation. For example, filtrate container **120** and the secondary container **128** are formed integrally with their respective lines **162**, **188**.

(31) On the other hand, lines **166**, **168**, **176**, **180**, **188** may be formed with an attachment site (such as an end formed to be sealed to the container or with a connector, such as a luer lock connector, attached thereto) to connect to the containers **110**, **122**, **124**, **126**, **114** at the time of use.

(32) The syringe **116** may be connected via a line **192** to a port **194**, and the syringe **118** may be connected via a line **196** to a port **198**. In a similar fashion, the spinning membrane **112** may be connected at an inlet of the spinning membrane **112** by a line **200** to an inlet port **202**, and at a first outlet via a line **204** to a first outlet port **208** and at a second outlet via a line **206** to a second outlet port **210**. In addition, an air vent port **212** is provided, and the air vent port **212** is connected to a

filter **214** via a line **216**. Because of the proximity of the spinning membrane **112**, the syringes **116**, **118**, and the filter **214** to the cassette **130**, one or more of the lines **192**, **196**, **200**, **204**, **208**, **216** may be defined by portions of cassette **130** itself.

(33) According to the illustrated embodiment, and as seen in FIGS. **2** and **4-6**, the cassette **130** includes a housing **220** defined by a frame **222** to which side walls **224**, **226** are attached (see FIGS. **2** and **7**). The walls **224**, **226** may be attached about the periphery of the frame **222**, as well as along structures of the frame that define the channels, clamps, and sensor stations discussed above. The walls **224**, **226** may be attached through the use of joining techniques, such as ultrasonic welding, or may be attached by holding the wall **224**, **226** and the frame **22** in contact with each other through the application of force.

(34) A negative pressure may be drawn on the side wall **224** of the cassette **130**. Drawing a negative pressure on the wall **224** of the cassette **130** is believed to prevent the collapse of the channels defined within the housing **220**. This is particularly important in a system that uses syringes **116**, **118** and syringe pumps **142**, **144** in that the syringe pumps operate, at least in part, by drawing negative pressures within the fluid paths defined, at least in part, by the channels. The application of negative pressure to the wall **224** of the cassette **130** compensates, at least in part, for the negative pressures drawn within the fluid paths.

(35) Turning next to FIG. **4**, it will be noted that the frame **222** defines the afore-mentioned plurality of separate and distinct channels, which channels may be connected to one of the ports discussed above. The channels may also have one or more apertures disposed at locations along the lengths of the channel. These apertures may be used to connect the channels, via the clamps or sensor stations, for example, to other channels. Together, the channels may define flow paths (or fluid paths, or fluid flow paths) between the containers **110**, **114**, **122**, **124**, **126**, **128**, syringes **116**, **118**, and the spinning membrane **112**.

(36) Starting at the left-hand side of the cassette **130**, a channel **230** is connected to the port **164**, and includes an aperture **270**. A channel **232** is connected to port **192**, and includes apertures **272**, **278**. A channel **234** is connected to port **212**, and includes aperture **274**. A channel **236** is connected to port **196**, and includes apertures **276**, **296**, **304**, **310**. A channel **238** includes an aperture **286**, while a channel **240** includes an aperture **280**. A channel **242** is connected to port **208**, and is connected to channel **240** via a station and unnumbered apertures of channels **240**, **242**; in a similar fashion, channel **238** is connected to channel **236** via a station and unnumbered apertures of channels **236**, **238**.

(37) Towards the middle of the cassette **130**, a channel **244** includes apertures **282**, **294**. A channel **246** is attached to port **178**, and includes aperture **284**. A channel **248** is attached to port **174**, and includes aperture **288**. A channel **250** includes apertures **290**, **292**, **302**. A channel **252** is connected to the inlet port **202** of the spinning membrane **112**, and is connected to the channel **250** via a station and unnumbered apertures of channels **250**, **252**.

(38) At the right-hand side of the cassette **130**, a channel **254** is connected to port **210**, and includes aperture **298**. On the other hand, at the right-hand side of the page, a channel **256** is connected to port **182** and includes an aperture **300**. A channel **258** is connected to port **190**, and includes an aperture **306**. Finally, a channel **260** is connected to port **186**, and includes an aperture **308**.

(39) As seen in FIG. **5**, each of the apertures included in the channels **236-260** is associated with one or more of the other apertures. In most instances, each aperture is associated with one of the other apertures; in one instance, three apertures are associated with each other. Each grouping of two or more apertures is associated with a chamber on the reverse side of the cassette **130** from that illustrated in FIG. **4**, which chamber then defines one of the clamps.

(40) In particular, apertures **270**, **272** are grouped, and define in part a selectable junction or clamp **320**, while apertures **274**, **276** are grouped, and define in part a clamp **321**. The apertures **278**, **280**, **282** are grouped, and define in part a clamp **322**. The apertures **284**, **286** define in part a clamp **323**, the apertures **288**, **290** define in part a clamp **324**, the apertures **292**, **294** define in part a clamp

325, and the apertures **296**, **298** define in part a clamp **326**. Finally, the apertures **300**, **302** define in part a clamp **327**, the apertures **304**, **306** define in part a clamp **328**, and the apertures **308**, **310** define in part a clamp **329**. The clamps **320-329** are also shown in dashed line in FIG. **6** with the markings of the apertures removed, for ease of illustration relative to the associations of the clamps **320-329** with the channels **230-260**.

(41) As mentioned above, each of the groupings of apertures is associated with a chamber, which chamber and the features thereof further define one of the clamps **320-329**. An exemplary clamp (for example, clamp **325**) is illustrated in larger scale in FIG. **7** so that the cooperation of the structures of the chamber may be visualized (the structures of the corresponding channels have been omitted for ease of illustration). While the clamp illustrated in FIG. **7** has only two apertures in cross-section, this structure also is applicable to those clamps that have more than two apertures.

(42) The clamp illustrated includes a chamber wall **340** that is formed as part of the frame **222**, and extends from a frame wall **342**. The chamber wall **240** encloses a circular region as viewed in FIG. **5**, and thus may also be described as a circumferential or peripheral wall. The side wall **226**, which may be made of a flexible material, is attached to an edge **344** of the chamber wall **340**, and with the chamber wall **340** and the frame wall **342** define an enclosed region or space **346**. The apertures **292**, **294** pass through the frame wall **342**, and one of the apertures **294** has a rim or flange **350** disposed about its circumference or periphery. The distance of an edge **352** of the rim **350** from the frame wall **342** is not as great at the distance of the edge **344** from the frame wall **342**.

(43) As illustrated in dashed line in FIG. **7**, a portion of the wall **226**, also referred to as a deflectable surface, may be brought into contact with the edge **352** of the rim **350** to cover the aperture **294** so that fluid cannot flow between the aperture **294** and the volume or space **346**. At the same time, the fluid flow between the aperture **292** and the space **346** may remain unobstructed because the deflectable surface does not cover the aperture **292**. However, by closing the aperture **294**, fluid flow may be interrupted along the fluid path defined by the channels **244**, **250** associated with the apertures **294**, **292**, respectively. The wall **226** may be selectively deflected to abut the rim **350** through the use of an actuator **354** coupled to the controller **150**, which actuator **354** may be defined in part by a shaft **356** that moves along an axis **358** (for example, where the shaft **356** is part of an electronic linear actuator). When the shaft **356** is advanced in the direction of the wall **342**, the shaft **356** deflects the wall **226** to abut the edge **352** and close the aperture **294**. When the shaft **356** is withdrawn away from the wall **342**, the wall **226** moves away from the edge **352** and the aperture **294** is open and in fluid communication with the space **346**.

(44) As illustrated in FIG. **8**, the cassette interface **146** may include a plurality of actuators **354**, each within a space that is intended to be aligned with one of the clamps **320-329**. The actuators **354** each may define one of a plurality of actuator stations **360-369** that corresponds to a respective one of the clamps **320-329**. Each of the actuators stations **360-369** may be coupled to the controller **150**, as illustrated in FIG. **3**, and the controller **150** may control the movement of the actuators **354** in the direction of and away from the cassette **130** when the cassette **130** is disposed or mounted on the cassette interface **146**. The controller **150** may operate the actuators **354** in conjunction with the desired process.

(45) According to an alternative embodiment of the cassette, where the channels are defined not by structures of the housing, but by tubing or lines disposed within the housing, the clamps may be defined by pinch valves. See U.S. Pub. No. 2017/0204371, which is incorporated herein by reference.

(46) The cassette **130** may also include a number of air sensor chambers **330-334** disposed at points along the periphery of the frame **222**. See FIG. **5**. The cassette **130** is to be used with air sensors that are associated with each of the air chambers **330-334** such that it is not necessary that the emitter and detector be disposed on opposite sides of the frame **222**. Instead, the emitter and detector can be disposed on the same side of the frame **222**, providing a so-called single-sided air sensor. This may be beneficial because there is no need to provide a door to close over the cassette

130, the door having either an emitter or a detector mounted thereon, as would be the case with a pass-through sensor where the emitter and detector must be disposed on opposite sides of the cassette **130**.

(47) The single-sided air sensors may be in the form of an ultrasonic sensor that emits controlled, timed pulses of ultrasonic energy into the chamber **330-334** and senses the response time of the “echo” of the emitted energy. The ultrasonic sensors may be part of the reusable hardware **104** of the system **100**, and thus may be mounted opposite the cassette **130** with the cassette interface **146**. The ultrasonic sensors may each be spring-biased (e.g., combined with a spring-loaded insert), such that the sensors will come into contact with a wall of each of the chambers **330-334** when the cassette **130** is mounted to the hardware **104**, and in particular the cassette interface **146**. The echo time is believed to change when the liquid enters the chamber.

(48) As an alternative, each of the chambers may have a window that permits an optical sensor to be used therewith, the window being translucent at least to light of a wavelength emitted by a light emitter associated with the sensor. A single-sided (reflectance-based) optical sensor may be used to determine other things than the presence of an air/fluid interface, such as cell concentrations in the fluid as well.

(49) A further alternative is illustrated in FIG. **9**. According to this embodiment (in which the cassette **130** is otherwise structurally identical to the cassette **130** of FIGS. **2** and **4-6**), the single-sided sensors **380, 382, 384, 386, 388** are capacitive sensors. Instead of being part of the reusable hardware **104** of the system **100**, the sensors **380, 382, 384, 386, 388** are attached to the cassette **130**, and are part of the disposable fluid circuit **102** instead. As illustrated in FIG. **9**, the sensors **380, 382, 384, 386, 388** are disposed on a wall **390, 392, 394, 396, 398** of the chambers **330-334**. The sensors **380, 382, 384, 386, 388** may be attached to an outer surface of the respective wall **390, 392, 394, 396, 398** of the respective chamber **330-334** with an adhesive, for example.

(50) Each of the sensors **380, 382, 384, 386, 388** is connected to one of a plurality of connector pads **400, 402, 404, 406, 408** that are disposed on the cassette **130**, in particular on an outer surface of the cassette **130**. The connector pads **400, 402, 404, 406, 408** may be connected to the sensors **380, 382, 384, 386, 388** by a lead or trace **410, 412, 414, 416, 418** that also may be formed on the outer surface of the cassette **130**. The connector pads **400, 402, 404, 406, 408** are configured to be contacted by a plurality of connectors mounted on the reusable hardware **104** of the system **100**. For example, the hardware **104** may include a plurality of spring-biased or spring-loaded connectors (e.g., pogo pins) that are configured to contact the connector pads **400, 402, 404, 406, 408**.

(51) The capacitive sensors **380, 382, 384, 386, 388** may be in the form of a multi-layer electrical elements, such as a flexible, or flex, circuit. The sensors **380, 382, 384, 386, 388** may monitor the electrical capacity of a known area and sense distortions within the electrostatic field in that area when other materials (other than air) are proximate to the area. In particular, the sensor may include two elements separated by a gap from each other, creating a structure having a capacitance that varies when either air or a fluid (e.g., fluid within the chamber **330-334**) is proximate to the gap. In this fashion, the sensor **380, 382, 384, 386, 388** may determine whether air or fluid is in the chamber **330-334**.

(52) An embodiment of the cassette **130** including capacitive sensors **380, 382, 384, 386, 388** to be used to determine the presence or absence of fluid in the chambers **330-334** may have one or more of the following advantages. Direct application of the sensors **380, 382, 384, 386, 388** to the cassette **130** limits or eliminates the effect that variations in dimensional tolerances may have on the sensor's detection abilities where the sensors must instead be brought into contact with the cassette **130**, and particularly the chambers **330-334**. Additionally, the spatial distribution of the chambers **330-334** over the cassette **130** means that the tolerances must be considered over a large area of the surface of the cassette **130** if the sensors must be brought into contact with the chambers **330-334**. By comparison, the connector pads **400, 402, 404, 406, 408** may be localized to a specific

region of the cassette **130**. Further, instead of having to configure the entire sensor to be adjustable to address tolerances in the manufacture of the cassette **130**, the embodiment of FIG. **9** permits the tolerances to be addressed through the design of the connector, which may be more easily achieved through the use of conventional spring-biased or spring-loaded connectors or other traditional connector technologies.

(53) While the afore-mentioned use of flexible capacitive sensors has been made with reference to sensing air-fluid interfaces in a cassette, such as the embodiments of the cassette **130** illustrated herein, it will be recognized that these capacitive sensors may be used with other equipment, and particularly medical equipment, where an air-fluid interface must be determined relative to fluid flowing in a disposable circuit that is associated with reusable hardware. For example, the technology may be used other separation technologies, such as an AMICUS® separator or an Autopheresis C® system, both of which are available from Fresenius Kabi USA, Lake Zurich, Ill.

(54) In addition to the cooperation between the cassette **130** and the cassette interface **146**, the disposable fluid circuit **102** and the reusable hardware **104** cooperates in other ways as well.

(55) FIG. **10** illustrates additional details of the spinning membrane separator **112**, for example. Preferably, spinning membrane **112** is a spinning membrane separator of the type described in U.S. Pat. Nos. 5,194,145 and 5,053,121, U.S. Provisional Patent Application No. 61/451,903 and International Pub. No. WO 2012/125470, which are incorporated herein by reference in their entirety. As discussed above, the spinning membrane separator **112** has one inlet **430** at least two outlet ports **432**, **434**. The outlet **432** of spinning membrane **112** receives the waste from the wash (i.e., a non-cellular component of the cellular suspension and wash medium from the spinning membrane separator) and is connected to line **204**. The spinning membrane **112** preferably includes a second outlet **434** that is connected to line **206** and receives the desired biological cell/fluid product (e.g., washed cells).

(56) FIGS. **11** and **12** illustrate an embodiment of a syringe pump that may be used with either the first or the second syringe **116**, **118** and as either the first and/or second syringe pump **142**, **144**.

(57) The syringe pump is configured to use a syringe **450** with a syringe barrel **452** (which may be made of cyclic olefin copolymer, or other materials such as may be inert, optically clear) and a piston or plunger head assembly **454**. The piston head assembly **454** is moveable (translatable) between a first end **456** and a second end **458** of the barrel **452**.

(58) The piston head assembly **454** includes the piston **460** and an infrared reflector **462**, which defines one part of a position detector **464**. According to the illustrated embodiment, the position detector **464** also includes a plurality of transmitter/sensor pairs **466**, **468**. According to the illustrated embodiment, the transmitters (or emitters) **466** may be in the form of infrared light emitting diodes, and the sensors **468** may be in the form of infrared sensors. According to other embodiments, the transmitters and sensors may use visible or ultraviolet light, for example. The transmitter/sensor pairs **466**, **468** are disposed along the length of the barrel **452** between the first end **456** and the second end **458**. The reflector **462** may be in the form of a reflective strip that is disposed about the perimeter of a rigid disc **470** that is attached opposite the piston **460**.

(59) In operation, the position detector **464** (which could be coupled to the controller **150**, for example) would use the interaction between the transmitter/sensor pairs **466**, **468** and the reflector **462** to determine the position of the piston head assembly **454** along the barrel **452**. In particular, light emitted from the transmitter **466** would be received by the sensor **468** (or would be received over a threshold amount) if the light contacts the reflector **462**. Otherwise, the light would not be received by the sensor **468** (or would not be received below the threshold amount). Depending on the amount of light received by the sensor **468**, a signal generated by the sensor **468** would vary. See, e.g., FIG. **13**, wherein the signal of the sensor **468** varies as the piston head assembly **454** is drawn to the second end **458**, with the sensor signal representative of first air, then piston O-ring, reflector **462** (corresponding to the peak in sensor output), piston, and finally fluid opposite the sensor **468**. Depending on the signals received from the individual transmitter/sensor pairs **466**,

468, the controller **150** may determine the position of the piston head assembly **454** along the barrel **452** between the first and second ends **456**, **458**.

(60) A vacuum/pressure source (e.g., a diaphragm pump) **472** is attached via line (e.g., tubing) **474** to the end **458** of the barrel **452**. The end **458** is otherwise closed, forming a first variable volume space **476** between the closed end **458** of the barrel **452** and the piston head assembly **454**. Filtered air may be pumped into and out of the space **476** to cause the piston head assembly **454** to move between the first and second ends **456**, **458** of the barrel **452**. The movement of the piston head assembly **454** causes a second variable volume space **478** to open between the piston head assembly **454** and the first end **456** to receive fluid (e.g., a cell product) into the barrel **452**. Compare FIGS. **11** and **12**. Fluid may be drawn into (or may enter into) and pushed or delivered from the space **478** according to the movement of the piston head assembly **454**.

(61) In operation, the piston head assembly **454** starts at a first position, such as is illustrated in FIG. **11**. The controller **150** causes the vacuum/pressure source to operate, and draw vacuum behind the piston head assembly **454** (i.e., in space **476**). As a consequence, the piston head assembly **454** moves in the direction of the end **458** (i.e., from the end **456** to the end **458**) and draws fluid into the space **478** (see FIG. **12**). The controller **150** may subsequently operate the vacuum/pressure source to pump pressurized air into the space **476**. This causes the piston head assembly **454** to move in the direction of the end **456** (i.e., from the end **458** to the end **456**) and push fluid from the space **478**.

(62) It will be recognized that the pneumatic control of filtered air in and out of the space **476** provides certain advantages over the use of a syringe with a plunger arm where one end of the barrel remains open to the surrounding environment. By leaving the barrel end open, materials could collect on an inner surface of the barrel wall, such that movement of the piston head between the ends could permit the materials on the inner surface to interact with the fluid on the other (i.e., wet-side) of the piston head. The use of filtered air in the space **476** to move the piston **460** reduces or eliminates this potential source of contaminants. Further, the position detector **464** permits very precise control of the operation of the syringe pump **450**. Other embodiments may use a mechanical or electro-mechanical mechanism to move the piston head **460**, however.

(63) FIGS. **14-16** illustrate an alternative embodiment of a syringe pump that may be used with either the first or the second syringe **116**, **118** and as either the first and/or second syringe pump **142**, **144**.

(64) The syringe pump is configured to use a syringe **490** with a wall defining a syringe barrel **492** (which may be made of cyclic olefin copolymer, or other materials such as may be inert, optically clear). Fluid may be drawn into a first end **494** of the syringe **490**/syringe barrel **492**, and may fill the barrel **492** to a second end **496** of the barrel **492**. According to the illustrated embodiment of the syringe **490**, the syringe lacks a piston or plunger head assembly moveable (translatable) between the first and second ends **494**, **496**; that is, the bore of the barrel **492** is without a piston or plunger disposed therein, or is plungerless. See FIGS. **15** and **16**. According to other embodiments, an optional piston or plunger head assembly may be included.

(65) The syringe **490** includes one or more capacitive sensors **498** that are configured to determine a location of an air/fluid interface (or, alternatively, the piston or plunger head assembly) within the barrel **492** of the syringe **490**. As illustrated, the sensor **498** is a single sensor. The sensor **498** is illustrated in solid line in FIG. **14**, and in dashed line in FIGS. **15** and **16**. The sensor **498** may be in the form of a flexible, or flex, circuit that is disposed on an outer surface **500** of the barrel **492**; for example, the sensor **498** may be attached to the outer surface of the barrel **492** using an adhesive.

(66) So that the position of the sensor **498** relative to volumes defined within the barrel **492** may be more easily determined in FIGS. **15** and **16**, the sensor **498** is not illustrated as extending from the first end **494** to the second end **496** of the barrel **492**. In practice, it would be preferred that the sensor **498** extend entirely from the first end **494** of the barrel **492** to the second end **496** of the barrel **492**. According to certain embodiments, however, there may be some amount of spacing

permitted at one or both of the ends **494**, **496** of the barrel **492**.

(67) The sensor **498** includes a pair of elements **502**, **504** with a gap **506** disposed therebetween. The sensor **498** monitors the electrical capacity and senses distortions within the electrostatic field when other materials (other than air) are proximate to the sensor **498**. As a consequence, the sensor **498** may determine the location of an air/fluid interface within the barrel **492** and/or the amount of fluid in the barrel **492**. To better facilitate this determination, it is preferred that the barrel **492** be disposed with the barrel **492** oriented substantially vertically with respect to gravity and with the first end **494** oriented downwardly. See, e.g., FIG. 1.

(68) A vacuum/pressure source (e.g., a diaphragm pump) **508** is attached via line (e.g., tubing) **510** to the end **496** of the barrel **492**. The source **508** is used to draw fluid into the syringe **490**, and to push the fluid from the syringe **490**. As illustrated in FIGS. 15 and 16, fluid has been drawn into the syringe **490** such that an air/fluid interface **512** exists between a first volume or space **514** of the syringe **490** that is filled with fluid, and a second volume or space **516** of the syringe **490** that is filled with air. Because the end **496** is otherwise closed (by a cap with a filter, for example), the volume of both spaces **514**, **516** may be varied by pumping air into and drawing air from the syringe **490**, and in particular the space **516**. The pumping of filtered air into and out of the space **516** causes the interface **512** to move between the first and second ends **494**, **496** of the barrel **492**.

(69) In operation, the interface **512** starts at a first position, such as may be slightly to the right of the position illustrated in FIG. 15. The controller **150** causes the vacuum/pressure source **508** to operate, and draw vacuum from the syringe **490** (i.e., from space **516**). As a consequence, the interface **512** moves in the direction of the end **496** (i.e., from the end **494** to the end **496**) as fluid is drawn into the space **514** (i.e., from the state illustrated in FIG. 15 to that illustrated in FIG. 16). The controller **150** may subsequently operate the vacuum/pressure source to pump pressurized air into the space **516**. This causes the interface **512** to move in the direction of the end **494** (i.e., from the end **496** to the end **494**) as fluid is pushed from the space **514** (i.e., from the state illustrated in FIG. 16 to that illustrated in FIG. 15).

(70) It will be recognized that the pneumatic control of filtered air in and out of the space **516** provides certain advantages over the use of a syringe with a plunger arm and associated piston or plunger head where one end of the barrel remains open to the surrounding environment, as mentioned above. By leaving the barrel end open, materials could collect on an inner surface of the barrel wall, such that movement of the piston between the ends could permit the materials on the inner surface to interact with the fluid on the other (i.e., wet-side) of the piston head. The use of filtered air in the space **516** to move an interface **512** similarly reduces or eliminates this potential source of contaminants. Moreover, in those embodiments where the piston or plunger head is omitted, any contamination attributable to the piston or plunger head itself may be prevented.

(71) The sensor **498** may be coupled, for example, to the controller **150** to provide the position/volume information to the system **100**. In this regard, the sensor **498** may be coupled to a pair of connector pads disposed on the outer surface **500** of the syringe **490**. These connector pads may be disposed at the second end **496** of the syringe **490**; alternative, the connector pads could be disposed elsewhere on the outer surface **500** of the syringe **490**. A pair of spring-biased or spring-loaded connectors (e.g., pogo pins) may be mounted on the hardware **104**, and may be configured to cooperate with the connector pads to couple the sensor **498** to the controller **150**.

(72) As was mentioned above relative to the air/fluid interface sensors **380**, **382**, **384**, **386**, **388**, a syringe **490** (which may be pistonless or plungerless) incorporating a capacitive sensor **498** may be used not only as part of the syringe pump of the system **100** described herein, but the syringe **490** with sensor **498** may be used in other medical devices or systems that include a syringe pump, such as fill and finish systems illustrated and described in U.S. Pub. No. 2018/0155070 or the small volume cell processing systems described in U.S. Pub. No. 2017/0204371 or U.S. Pat. App. Pub. No. 2019/0351113, all of which are incorporated herein by reference.

(73) Having discussed the structure of the illustrated embodiments of the fluid circuit **102** and the

corresponding equipment of the reusable hardware **104**, the operation of the system **100** is now discussed with reference to FIG. **17**. As much of the operation of the system **100** involves control of the fluid flow between the containers **110**, **114**, **120**, **122**, **124**, **126**, **128**, the syringes **116**, **118**, and the spinning membrane **112** through the cassette **130**, reference is also made to FIGS. **1**, **2** and **6**.

(74) A method **530** of operating the system **100** may begin with one or more pre-processing actions at blocks **532**, **534**, **536**. While each block **532**, **534**, **536** describes a different general type of activity, the actions at blocks **532**, **534**, **536** may include a plurality of individual actions. For example, priming the circuit **102** at block **536** may include individual actions of priming different portions of the circuit **102**, but for ease of illustration, the actions have been grouped together at block **536**.

(75) Starting then with block **532**, the circuit **102** is installed on the hardware **104**. The controller **150** may determine when this is complete by checking one or more sensors, or the controller **150** may wait for an input to be received from the user via an input device, such as a touch screen. Once the controller **150** has determined that the circuit **102** is installed, the method passes to block **534**.

(76) At block **534**, the controller **150** may perform numerous tests on the circuit **102** before any fluid is added to the circuit **102**. Certain tests, or checks, are performed on the syringe pumps **142**, **144**, other checks are performed on the clamps **320-329**. Because no fluid has been introduced to the circuit **102**, these checks may be referred to as dry checks. After the checks have been performed, the containers **122**, **124**, **126**, **128** containing the solutions for the process to be performed using the system may be connected to the circuit **102**, and in particular to the lines **166**, **168**, **176**, **180**.

(77) The first set of actions may be performed on the syringes **116**, **118** and the syringe pumps **142**, **144**, and in particular those embodiments of the syringe pumps **142**, **144** including syringes with piston or plunger head assemblies and an optical sensor arrangement. As such, reference is made to the structures of the embodiment of FIGS. **11** and **12**. Initially, the pistons (e.g., **460**) of the syringes **116**, **118** are drawn to the second end of the syringe (e.g., end **458**, which end may be referred to as the upper end with the syringes **116**, **118** in the orientation of FIG. **1**) and then to the first end of the syringe (e.g., end **456**, which end may be referred to as the lower end with the syringes **116**, **118** in the orientation of FIG. **1**), which is done to permit sensor normalization to be conducted (typically either when the piston is at the second end or the first end). During these actions, all of the clamps **320-329** are left open. A circuit (or kit) integrity check is then performed by moving the pistons of the syringes **116**, **118** to the second end of the syringe with only clamps **320**, **321** closed. If the piston of either syringe **116**, **118** is able to move toward the second end, then this suggests a leak in the circuit **102** or that the clamp **321** is unable to maintain a vacuum.

(78) After the first set of actions, further checks are performed on the clamps **320-329**. For example, clamps **322** and **325** may be closed, while the remaining clamps are open, and the piston of syringe **116** is moved toward the second end. Clamp **320** is then closed after a period of time, which should cause the piston in syringe **116** to stop moving and a negative pressure to build in the syringe **116**. A similar process can be conducted for other clamps. For example, clamps **321**, **322**, **326** can be tested with all clamps except clamps **321**, **322**, and **326** closed, and then each of clamp **321**, **322**, **326** closed some time after the piston of syringe **116** is moved toward the second end. For clamp **329**, all clamps **320-328** are closed and the piston of syringe **118** is moved toward the second end. The clamp **320** may also be checked again using the second syringe pump **144** and the second syringe **118**, with the process being generally the same except that clamps **320**, **322**, **325**, **326** are left open, and the piston of syringe **118** is advanced towards its second end.

(79) Once the checks have been performed, the method **530** continues with the containers **110**, **122**, **124** and potentially container **126** being attached at the end of the actions of block **534** or the beginning of block **536**. With the containers **110**, **122**, **124** attached, the circuit **102** may be primed at block **536**.

(80) The priming of the circuit **102** may start with the priming of the fluid path to the second syringe pump **144**. To do this, the controller **150** may open clamps **324, 326** (remainder closed) and cause the piston of the second syringe **118**, in those embodiments that include a piston, to move toward the second end. In this and the remaining actions described below, where a piston is not present, this may be read instead as a reference to the movement of the air/fluid interface instead. This action draws wash fluid from the container **122, 124** through the port **174** and the channels **248, 250, 252** into the inlet **430** of the spinning membrane **112**. The fluid passes through the spinning membrane **112**, through port **210** and channels **236, 254** into the port **196** and syringe **118**.

(81) The fluid in the syringe **118** may be used to prime the path between the syringe pump **144** and the vent port **212** by closing all clamps except **321**. The piston of the syringe **118** is then moved toward the first end to force fluid from the syringe **118** into channels **236, 234**.

(82) The priming of the circuit **102** may continue with the priming of the fluid path to the first syringe pump **142**. To do this, the controller **150** opens clamps **322, 324** (remainder closed) and causes the piston of the first syringe **116** to move toward the second end. This draws wash fluid from the container **122, 124** through the port **174** and the channels **248, 250, 252** into the inlet **430** of the spinning membrane **112**. The fluid passes through the spinning membrane **112**, through port **208** and channels **232, 240, 242** and into the port **192** and syringe **116**.

(83) The fluid drawn into the first syringe **116** may be used to prime the fluid path to the source container **110**. The controller **150** opens clamps **325, 327** and causes the piston of the first syringe **116** to move toward the first end. This pushes wash fluid from the syringe **116** through the port **192** and through channels **232, 244, 250, 256** to port **182**. The fluid is able to pass from channel **232** to channel **244** because of the multiple apertures in clamp **322** that remain open to the chamber associated with that clamp even when the clamp **322** is closed. The fluid is pushed from the port **182** along the line **180** and into source container **110**. This priming step removes air from the line **180** such that the system **100** is ready to begin processing cells.

(84) The method **530** continues at block **538** with the controller **150** operating the system **100** to perform a procedure according to a protocol on the fluid in the source container **110**. As one example, the controller **150** may operate the system **100** to separate cells from the fluid in the container **110**, rinse the container **110** and wash the cells, and then pass the washed cells to container **128** for additional processing. As was the case with the actions at blocks **252, 254, 256**, the actions at block **258** may include numerous individual actions, at least some of which may be repeated according to the amount of fluid in the source container **110**, for example.

(85) In this regard, a further flowchart is provided in FIG. **18** to illustrate the actions of the block **258** in FIG. **17**. In general terms, the separation of the cells from the fluid in the container **150** requires that the cells be transferred from the container **110** to the spinning membrane **112**, the spinning membrane **112** be operated to separate the cells from the filtrate, the filtrate transferred first to the syringe **116** and then to the container **120**, and the cells transferred first to the syringe **118** and to the container **128**, at least according to the illustrated embodiment. Certain of the actions illustrated in FIG. **18** follow in a necessary order, for example, operating the syringe pump **142** occurs as a consequence of a determination that the syringe **116** is full. On the other hand, certain actions may be performed in any order; for example, the determination if the syringe **116** is full may follow the determination if the spinning membrane **112** is full, rather than the order illustrated. Some of the actions may be optional, and an attempt has been made to represent optional actions with the use of dashed line.

(86) To perform the separation of the cells from the fluid in the container **110** at block **550**, clamps **322, 327** may be opened (remainder closed) and the piston of syringe **116** is moved toward the second end. This draws fluid from the source container **110** into the port **182**, channels **250, 252, 256** and into inlet **430** of the spinning membrane **112**. Fluid is drawn from the spinning membrane **112** through ports **208** and channels **232, 240, 242** into the port **192** and the syringe **116**. While the fluid and cells are flowing into the spinning membrane **112**, a rotor **436** of the spinning membrane

112 rotates at a separation rate defined by the protocol, and fluid is drawn from the spinning membrane **112** while the target cells accumulate in an annulus **438** of the spinning membrane **112** between an outer housing **440** of the spinning membrane **112** and an outer surface **442** of a membrane **444**. See FIG. **10**. The cells accumulate in the annulus **438** because the clamp **326** is closed.

(87) Depending on the amount of fluid present in the container **110**, it may be necessary to empty the syringe **116** from time to time into the filtrate container **120**. This may be done when the syringe **110** is full or reaches a certain threshold volume, as determined at block **552**. This may also be done when the system needs to reset the position the filtrate syringe piston at the first end (e.g., end **456**), for example. At block **554**, the controller **150** opens only clamp **320** while causing the piston of the syringe **116** to move in the direction of the first end, causing fluid to flow along the fluid path defined by channels **230**, **232**, port **164**, and line **162** into the container **120**.

(88) Depending on the accumulation of the cells in the annulus **438** of the spinning membrane **112**, it may be desirable to wash the cells in the spinning membrane **112** and move the cells to the syringe **118** (which may be referred to as harvesting the cells). See blocks **556**, **558**, **560**. If the capacity of the spinning membrane **112** is not reached before the source container **110** is emptied (as determined at block **562**, for example), then the cell wash may be conducted after other actions have occurred, namely the rinsing of the source container **110**. See blocks **564**, **566**, **568**, **570**, **572**. It is possible for the cell wash and harvest to be performed multiple times (e.g., once when the capacity of the spinning membrane **112** is reached at blocks **560**, **562** and once after the source container **110** is emptied at block **570**, **572**). According to certain embodiments, the cell wash at blocks **558**, **570** may not be performed, and the source rinse at blocks **566**, **568** may also be optional.

(89) To perform a cell wash (block **558**, **570**), the controller **150** opens the clamps **322**, **324** (remainder closed) and controls the piston of the syringe **116** to draw fluid from the containers **122**, **124** through the port **174** and channels **248**, **250**, **252**, through the spinning membrane **112**, port **208**, and channels **232**, **240**, **242** and into the syringe **116**. This causes the fluid originally surrounding the cells (which may be referred to as original supernatant) to be replaced with new solution (i.e., the wash media). To harvest the cells (block **560**, **572**), the controller **150** leaves clamp **324** open, closes clamp **322**, and opens clamp **326**. The piston of syringe **118** is caused to move toward the second end to draw fluid from the containers **122**, **124** through the port **174** and channels **248**, **250**, **252**, through the spinning membrane **112**, port **210**, and channels **236**, **254** and into the syringe **118**.

(90) When the source container **110** is emptied (block **562**), the controller **150** may operate the system **100** to rinse the source container **110** (block **566**), after which a cell wash and harvest is performed. To perform the rinse (block **568**), the controller **150** first empties the syringe **116**. Then, the controller **150** closes the clamp **320**, and opens clamps **324**, **325** (remainder closed) and moves the piston of the syringe **116** to draw fluid from the containers **122**, **124** through port **174** and channels **232**, **244**, **248**, **250** and into port **192** and syringe **116**. Once a volume of the wash media has been drawn (or loaded) into the syringe **116**, the syringe **116** is ready to deliver the rinse to the source container **110**. To do so, the controller **150** leaves clamp **325** open, closes clamp **324**, and opens clamp **327**. The controller **150** then causes the piston of syringe **116** to push fluid through channels **232**, **244**, **250**, **256** and port **182** into the source container **110**. With the wash media transferred to the source container **110**, the controller **150** can conduct a wash and harvest (blocks **570**, **572**).

(91) At this point (block **574**) or earlier if the syringe **118** is determined to be full (see blocks **576**, **578**), the contents of the syringe **118** may be transferred to the product container(s) **114**. See also, block **540** of FIG. **17**. Alternatively, according to the illustrated embodiment, the contents of the syringe **118** may be transferred to the container **128** at blocks **574**, **578**, where the concentration of the cells may be modified and other components may be added at blocks **542**, **546**, after which the

product containers are filled at block **540**. According to the illustrated embodiment, the component added may be a cryopreservation agent (or CPA).

(92) To begin, the contents of the syringe **118** are transferred to the container **128**. The controller **150** opens only clamp **329**, and causes the piston of the syringe **118** to move in the direction of the first end. This pushes the contents of the syringe **118** through channels **236**, **260** and port **186** into line **184** and container **128**. The controller **150** then closes the clamp **329** and opens clamps **324**, **326**, and draws fluid into the syringe **118** through port **174** from containers **122**, **124**, into channels **248**, **250**, **252**, the spinning membrane **112**, and channels **236**, **254**, and to the port **196**. The controller then closes clamps **324**, **326**, opens clamp **329**, and causes the piston of syringe **118** to move in the opposite direction to push the contents (wash media) from the syringe **118** into container **128**.

(93) According to the illustrated embodiment, the controller **150** may pause the method **530** at block **544**. In fact, the controller **150** may pause the method **530** twice: once to permit a sample to be drawn from the container **128**, and a second time to permit the container **126** to be connected to the circuit **102** if the container **126** was not attached previously. Once the desired activities have been conducted, the method continues with the addition of the CPA.

(94) As part of the addition of the CPA to the container **128**, the controller **150** may first open only clamp **323** and cause the syringe **118** to draw CPA solution from the container **126** via line **176** and port **178** and through channels **236**, **238**, **246** into the syringe **118**. At this point, the clamp **323** is closed, and clamp **321** may be opened to permit the syringe **118** to push any excess air from the syringe **118** and out the vent port **212** and filter **214** via channels **234**, **236**. Preferably, enough air is left in the syringe **118** to have an air chase from the second syringe pump **144** to the container **128**. The controller **150** then closes clamp **321**, opens clamp **323** and draws the desired volume of CPA solution from container **128** via line **176** and port **178** and through channels **236**, **238**, **246** into the syringe **118**. The controller **150** closes clamp **323**, opens clamp **329** and moves the piston of syringe **118** to push the CPA solution from the syringe **118** into the container **126**.

(95) As illustrated, the embodiment of the system **100** includes a table **600** (see FIGS. **1** and **3**) on which the container **128** is disposed to oscillate therewith. The table **600** may be mounted on a motor-drive shaft **602** that permits the table **600** to oscillate about an axis **604**. The controller **150** may control the table **600** (via the motor) to cause the table **600** to oscillate to agitate the contents of the container **128**, encouraging mixing of the contents. This agitation may be performed, for example, while the CPA solution is being added to the container **128**. The agitation may be continued for an additional time after the CPA solution has been added to encourage adequate mixing. The table **600** may also include a cooling or heating element that permits the material in the container **128** to be maintained at a particular temperature.

(96) According to certain embodiments, the container **128** may be detached from the circuit **102**. However, according to the illustrated embodiment, the contents of the container **128** are instead filled into one or more product containers **114** that are attached to the circuit **102**. The system **100** may include a scale **606** (see FIGS. **1** and **3**) for weighing the contents of the container(s) **114**, although the sensitive nature of the volume control on the syringe pump **144** makes the use of the scale **606** more in the nature of a secondary check.

(97) To begin the fill activity, the controller **160** opens the clamp **329** and causes the syringe pump **144** to draw fluid from the container **128** via line **184** and port **186** into channels **236**, **260** to prime the fluid path between the container **126** and the pump **144**. The controller **150** then closes the clamp **329** and opens clamp **321** to vent excess air from the syringe **118** via channels **234**, **236**. Preferably, an air chase volume remains in the syringe **118** if the final dose volume and air chase volume can be delivered in one syringe stroke. The air chase volume should be sufficient to fully move the final dose volume from the syringe **118** to the final container **114**.

(98) The controller **150** then closes clamp **321** and opens clamp **329** and causes the syringe pump **144** to draw fluid from the container **128** via line **184** and port **186** into channels **236**, **260** to fill the

syringe **118** with the desired volume of product. The controller **150** then closes clamp **329**, opens clamp **328**, and causes the syringe pump **144** to push fluid from the syringe **118** to one of the containers **114**. The controller **150** may close the clamp **328** and open the clamp **321** to permit air to be drawn into the syringe **118**, which air is then pushed from the syringe **118** with the clamp **321** closed and the clamp **328** open to provide an air chase volume to force the product solution into the container **114**.

(99) The system **100** may include other equipment as part of the hardware **104**, in addition to the equipment already discussed, as illustrated in FIGS. **1** and **3**. For example, the system **100** may include a display **608** with touch screen **610** to permit information to be entered into the system, including information regard the protocol of the procedure to be carried out by the system **100**. The display **608** may be an electronic display, for example, with the touch screen **610** mounted thereon. Other input devices may be included, such as a pointer (e.g., mouse) and keyboard or keypad. As illustrated in FIGS. **1** and **3**, an input device in the form of a barcode reader **612** may be attached to the system **100** to permit information to be inputted into the system **100** (and the controller **150**) by scanning or reading a barcode, such as may be applied to the fluid circuit **102** or one or more of the containers **110**, **114**, **120**, **122**, **124**, **126**, **128**. Other output devices also may be included, such as one or more lights (e.g., light emitting diodes or bulbs) **614**, which may be used to signal alerts, events or machine states to the operator.

(100) The system **100** may also be used with one or more ancillary or secondary devices or peripherals, which peripherals may include valves, pumps, etc. to be used to control the filling of the product containers, for example. The system **100** (and in particular, the controller **150**) may be in communication with the peripheral(s), and data may be transmitted back and forth between the system **100** and the peripheral(s) or may be shared between the system **100** and the peripheral(s). In fact, the peripheral(s) may have its own controller (as illustrated, which controller may include a microprocessor, other circuits or circuitry and one or more memories, which may be one or more tangible non-transitory computer readable memories, with computer executable instructions by which the microprocessor is programmed and which when executed by the microprocessor may cause the microprocessor to carry out one or more actions being stored on the memory/memories) that is in communication with the controller **150**.

(101) According to at least one embodiment, the ancillary or secondary device may be in the form of an external array of valves that can control the passage of fluid between the fluid circuit and the product containers, and the state of the valves may be controlled or triggered by the controller **150** via communication between the controller **150** and the controller associated with the array of valves

(102) Thus, an improved method and system for processing (e.g., concentrating or washing) small volumes of biological cells has been disclosed, in conjunction with an improved method and system for filling low-volume containers with the small volumes of processed. The description provided above, and the other aspects provided below, are intended for illustrative purposes, and are not intended to limit the scope of the disclosure to any particular method, system, apparatus or device described herein.

OTHER ASPECTS

(103) Aspect 1. A fluid processing system comprising: a disposable fluid circuit comprising: a separation chamber; and a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber; and reusable hardware configured to accept the disposable fluid circuit and comprising: a separator connected to the separation chamber; a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor; and at least one controller

coupled to the separator, the at least one actuator and the capacitive sensor via the coupling, the controller configured to selectively operate the separator and the at least one actuator to provide a procedure according to a protocol.

(104) Aspect 2. The fluid processing system according to aspect 1, wherein the at least one capacitive sensor comprises a flexible circuit.

(105) Aspect 3. The fluid processing system according to aspect 1 or 2, wherein the at least one capacitive sensor comprises a connector pad, and the coupling comprises a spring-biased connector.

(106) Aspect 4. The fluid processing system according to aspect 3, wherein the spring-biased connector comprises a pogo pin.

(107) Aspect 5. The fluid processing system according to any one of aspects 1-4, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.

(108) Aspect 6. The fluid processing system according to any one of aspects 1-5, wherein: the disposable fluid circuit comprises first and second syringes, and the reusable hardware comprises first and second syringe pumps, the first and second syringes coupled to the first and second syringe pumps respectively, the first syringe pump configured to move fluid into and out of the first syringe and the second syringe pump configured to move fluid into and out of the second syringe.

(109) Aspect 7. The fluid processing system according to any one of aspects 1-6, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.

(110) Aspect 8. The fluid processing system according to any one of aspects 1-4, wherein: the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator; the disposable fluid circuit comprises first and second syringes, and the reusable hardware comprises first and second syringe pumps, the first and second syringes coupled to the first and second syringe pumps respectively, the first syringe pump configured to move fluid into and out of the first syringe and the second syringe pump configured to move fluid into and out of the second syringe; the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane, the filtrate outlet and the first syringe, a second path between the first syringe and a filtrate container, and a third path between a wash media container, the spinning membrane inlet, the retentate outlet and the second syringe; and the at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the capacitive sensor via the coupling, the controller configured to selectively operate the spinning membrane separator drive, the first and second syringe pumps, and the at least one actuator to provide a procedure according to a protocol.

(111) Aspect 9. The fluid processing system according to aspect 8, wherein: the first syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the first syringe to draw fluid into and push fluid from the first syringe; and the second syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the second syringe to draw fluid into and push fluid from the second syringe.

(112) Aspect 10. A fluid processing system comprising: a disposable fluid circuit comprising: a separation chamber; at least one plungerless syringe, the syringe comprising a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe; and a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels

connected to the first end of the barrel of the at least one plungerless syringe; and reusable hardware configured to accept the disposable fluid circuit and comprising: a separator connected to the separation chamber; at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe; a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor; and at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the controller configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

(113) Aspect 11. The fluid processing system according to aspect 10, wherein the at least one capacitive sensor comprises a single capacitive sensor disposed on the outer surface of the at least one syringe from the first end to the second end of the barrel.

(114) Aspect 12. The fluid processing system according to aspect 10 or 11, wherein the capacitive sensor comprises a flexible circuit.

(115) Aspect 13. The fluid processing system according to any one of aspects 10-12, wherein the capacitive sensor comprises a connector pad, and the coupling comprises a spring-biased connector.

(116) Aspect 14. The fluid processing system according to aspect 13, wherein the spring-biased connector comprises a pogo pin.

(117) Aspect 15. The fluid processing system according to any one of aspects 10-14, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.

(118) Aspect 16. The fluid processing system according to any one of aspects 10-15, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.

(119) Aspect 17. The fluid processing system according to any one of aspects 10-16, wherein: the at least one syringe comprises a first syringe and a second syringe, and at least one syringe pump comprises a first syringe pump and a second syringe pump, the second end of the barrel of the first syringe coupled to the first syringe pump, the first syringe pump configured to draw fluid into and push fluid from the first syringe, and the second end of the barrel of the second syringe coupled to the second syringe pump, the second syringe pump configured to draw fluid into and push fluid from the second syringe.

(120) Aspect 18. The fluid processing system according to aspect 17, wherein: the first syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the first syringe to draw fluid into and push fluid from the first syringe; and the second syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the second syringe to draw fluid into and push fluid from the second syringe.

(121) Aspect 19. The fluid processing system according to aspect 18, wherein: the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the reusable hardware comprises a spinning membrane separator drive coupled to the spinning membrane separator; wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane, the filtrate outlet and the first syringe, a second path between the first syringe and a filtrate container, and a third path between a wash media container, the spinning membrane inlet, the retentate outlet and the second syringe; and the at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the capacitive sensor via the coupling, the controller configured to selectively operate the spinning membrane separator drive, the first and second

syringe pumps, and the at least one actuator to provide a procedure according to a protocol.

(122) Aspect 20. A fluid processing system comprising: a disposable fluid circuit comprising: a separation chamber; at least one plungerless syringe, the syringe comprising a wall defining a barrel having a first end and a second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the syringe; and a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe, at least one interface sensor chamber in fluid communication with at least one of the plurality of separate channels, the at least one interface sensor chamber defined at least in part by a wall, and at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber; and reusable hardware configured to accept the disposable fluid circuit and comprising: a separator connected to the separation chamber; at least one syringe pump, the second end of the barrel of the at least one syringe coupled to the at least one syringe pump, the at least one syringe pump configured to draw fluid into and push fluid from the at least one syringe; a coupling connected to the at least one capacitive sensor disposed on an outer surface of the syringe; a control cassette interface having at least one actuator for each of the selectable junctions and a coupling connected to the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber; and at least one controller coupled to the separator, the at least one syringe pump and the control cassette interface, the controller configured to selectively operate the separator, the at least one syringe pump and the interface to provide a procedure according to a protocol.

(123) Aspect 21. The fluid processing system according to aspect 20, wherein the at least one capacitive sensor disposed on an outer surface of the syringe and the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber each comprises a flexible circuit.

(124) Aspect 22. The fluid processing system according to aspect 20 or 21, wherein the at least one capacitive sensor disposed on an outer surface of the syringe and the at least one capacitive sensor disposed on the wall of the at least one interface sensor chamber each comprises a connector pad, and the couplings each comprises a spring-biased connector.

(125) Aspect 23. The fluid processing system according to aspect 22, wherein the spring-biased connector comprises a pogo pin.

(126) Aspect 24. The fluid processing system according to any one of aspects 20-23, wherein the at least one capacitive sensor disposed on an outer surface of the syringe comprises a single capacitive sensor disposed on the outer surface of the syringe from the first end to the second end of the barrel.

(127) Aspect 25. The fluid processing system according to any one of aspects 20-24, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.

(128) Aspect 26. The fluid processing system according to any one of aspects 20-25, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.

(129) Aspect 27. The fluid processing system according to any one of aspects 20-26, wherein: the at least one syringe comprises a first syringe and a second syringe, and at least one syringe pump comprises a first syringe pump and a second syringe pump, the second end of the barrel of the first syringe coupled to the first syringe pump, the first syringe pump configured to draw fluid into and push fluid from the first syringe, and the second end of the barrel of the second syringe coupled to the second syringe pump, the second syringe pump configured to draw fluid into and push fluid from the second syringe.

(130) Aspect 28. The fluid processing system according to aspect 27, wherein: the first syringe

pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the first syringe to draw fluid into and push fluid from the first syringe; and the second syringe pump comprises a vacuum/pressure source, the vacuum/pressure source pumping filtered air into and out of the second syringe to draw fluid into and push fluid from the second syringe.

(131) Aspect 29. The fluid processing system according to aspect 28, wherein: the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the reusable hardware comprises a spinning membrane separator drive coupled to the spinning membrane separator; wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane, the filtrate outlet and the first syringe, a second path between the first syringe and a filtrate container, and a third path between a wash media container, the spinning membrane inlet, the retentate outlet and the second syringe; and the at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the capacitive sensor via the coupling, the controller configured to selectively operate the spinning membrane separator drive, the first and second syringe pumps, and the at least one actuator to provide a procedure according to a protocol.

Claims

1. A fluid processing system comprising: a disposable fluid circuit comprising: a separation chamber; at least one plungerless syringe, the at least one plungerless syringe oriented vertically and comprising a wall defining a barrel having a first end and a second end with the first end being located below the second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the at least one plungerless syringe; a flow control cassette comprising a housing containing a plurality of separate channels connected to a plurality of selectable junctions, at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe; the flow control cassette further comprising at least one separately defined interface sensor chamber in fluid communication with at least one of the plurality of separate channels connected to the first end of the barrel of the at least one plungerless syringe, the at least one separately defined interface sensor chamber further defined at least in part by a wall separate from the at least one of the plurality of separate channels, and at least one capacitive sensor disposed on the wall of the at least one separately defined interface sensor chamber to detect the presence of air; and reusable hardware configured to accept the disposable fluid circuit and comprising: a separator connected to the separation chamber of the disposable fluid circuit; at least one vacuum/pressure air source coupled to the second end of the barrel of the at least one plungerless syringe of the disposable fluid circuit, the at least one vacuum/pressure air source configured to pump filtered air into and out of the at least one plungerless syringe so as to form at least one syringe pump configured to draw fluid into and push fluid from the at least one plungerless syringe and selectively through the plurality of separate channels connected to the plurality of selectable junctions within the flow control cassette; a control cassette interface being coupled to the at least one capacitive sensor disposed on the wall of the at least one separately defined interface sensor chamber when the flow control cassette of the disposable fluid circuit is installed on the reusable hardware, with the control cassette interface having at least one actuator for each of the selectable junctions of the flow control cassette, and being coupled to the at least one capacitive sensor disposed on the outer surface of the wall of the at least one plungerless syringe; and at least one controller coupled to the separator, the at least one vacuum/pressure air source and the control cassette interface; the at least one controller configured to selectively operate the separator, the at least one vacuum/pressure air source and the control cassette interface to process a fluid according to a protocol.

2. The fluid processing system according to claim 1, wherein the at least one capacitive sensor on the outer surface of the wall of the at least one plungerless syringe comprises a single capacitive sensor disposed on the outer surface of the at least one plungerless syringe from the first end to the second end of the barrel.
 3. The fluid processing system according to claim 1, wherein the capacitive sensor on the outer surface of the wall of the at least one plungerless syringe comprises a flexible circuit.
 4. The fluid processing system according to claim 1, wherein the at least one capacitive sensor disposed on the wall of the at least one separately defined interface sensor chamber comprises a flexible circuit.
 5. The fluid processing system according to claim 1, wherein the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the separator comprises a spinning membrane separator drive coupled to the spinning membrane separator.
 6. The fluid processing system according to claim 1, wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining a plurality of fluid paths.
 7. The fluid processing system according to claim 1, with the disposable fluid circuit further comprising: a second plungerless syringe, the second plungerless syringe oriented vertically and comprising a wall defining a barrel having a first end and a second end with the first end being located below the second end, the barrel having a bore without a plunger disposed therein, and at least one capacitive sensor disposed on an outer surface of the wall of the second plungerless syringe, and with the reusable hardware further comprising: a second vacuum/pressure air source coupled to the second end of the barrel of the second plungerless syringe of the disposable fluid circuit, the second vacuum/pressure air source configured to pump filtered air into and out of the second plungerless syringe so as to form a second syringe pump configured to draw fluid into and push fluid from the second plungerless syringe and selectively through the plurality of separate channels connected to the plurality of selectable junctions within the flow control cassette.
 8. The fluid processing system according to claim 7, wherein: the separation chamber is defined by a spinning membrane separator having an inlet, a retentate outlet, and a filtrate outlet, and the reusable hardware comprises a spinning membrane separator drive coupled to the spinning membrane separator; wherein the plurality of separate channels are connected at the plurality of selectable junctions, each of the selectable junctions selectively connects at least two of the plurality of channels, and the channels and selectable junctions defining at least a first path between a source container, the spinning membrane separator, the filtrate outlet and the first plungerless syringe, a second path between the first plungerless syringe and a filtrate container, and a third path between a wash media container, the spinning membrane separator inlet, the retentate outlet and the second plungerless syringe; and the at least one controller coupled to the spinning membrane separator drive, the first and second syringe pumps, the at least one actuator, and the at least one capacitive sensor on the outer surface of the wall of the first plungerless syringe; the at least one controller configured to selectively operate the spinning membrane separator drive, the first and second syringe pumps, and the at least one actuator to provide a procedure according to a protocol.
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